PHYSICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

26. Consider a parallel plate capacitor of area A (of each plate) and separation 'd' between the plates. If E is the electric field and ε_0 is the permittivity of free space between the plates, then potential energy stored in the capacitor is

(1) $\frac{1}{4}\varepsilon_0 E^2 Ad$ (2) $\frac{1}{2}\varepsilon_0 E^2 Ad$

(3) $\varepsilon_0 E^2 Ad$ (4) $\frac{3}{4}\varepsilon_0 E^2 Ad$

Answer (2)

Sol. We know energy density

$$\rho_{av} = \frac{1}{2}\varepsilon_0 E^2$$

So potential energy = $\rho_{av} \times$ Volume

$$\Rightarrow \text{P.E.} = \frac{1}{2}\varepsilon_0 E^2 Ad$$

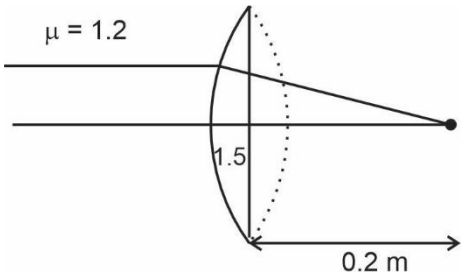
27. A thin plano convex lens made of glass of refractive index 1.5 is immersed in a liquid of refractive index 1.2. When the plane side of the lens is silver coated for complete reflection, the lens immersed in the liquid behaves like a concave mirror of focal length 0.2 m. The radius of curvature of the curved surface of the lens is

(1) 0.20 m (2) 0.15 m

(3) 0.25 m (4) 0.10 m

Answer (4)

Sol.



We have shown extrapolated figure for reflecting side as it went undeviated.

$$\frac{1.5}{v_1} - \frac{1.2}{\infty} = \frac{(1.5-1.2)}{R}$$

$$\frac{1.2}{0.2} - \frac{1.5}{v_1} = \frac{(1.2-1.5)}{-R}$$

$$\frac{1.2}{0.2} = \frac{0.3}{R} + \frac{0.3}{R}$$

$$\Rightarrow 6 = \frac{2\times 3}{10R} \Rightarrow R = \frac{1}{10}\text{m} = 0.10\text{ m}$$

28. For an experimental expression $y = \frac{32.3\times 1125}{27.4}$, where all the digits are significant. Then to report the value of y we should write

(1) $y = 1330$

(2) $y = 1326.2$

(3) $y = 1326.19$

(4) $y = 1326.186$

Answer (1)

Sol. $y = \frac{32.3\times 1125}{27.4} = 1326.18$

So we need to report to three significant digit.

So, y = 1330



29. A satellite is launched into a circular orbit of radius 'R' around the earth. A second satellite is launched into an orbit of radius 1.03R. The time period of revolution of the second satellite is larger than the first one approximately by

(1) 9%

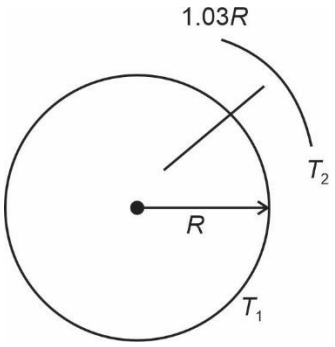
(2) 2.5%

(3) 3%

(4) 4.5%

Answer (4)

Sol.



$T_1 \propto (R)^{3/2}$

and $T_2 \propto (1.03R)^{3/2}$

$\Rightarrow \quad T_2 = (1.03R)^{3/2} \cdot T_1 \approx 1.045T_1$

So T_2 will larger by 4.5% w.r.t. T_1.

30. The amount of work done to break a big water drop of radius 'R' into 27 small drops of equal radius is 10 J. The work done required to break the same big drop into 64 small drops of equal radius will be

(1) 10 J

(2) 20 J

(3) 5 J

(4) 15 J

Answer (4)

Sol. $W_1 = s\,4\pi\left(\frac{R}{3}\right)^2 \times 27 - 4\pi R^2 \cdot s$

$\Rightarrow \quad W_1 = 4\pi R^2 \cdot s(2) = 10$ Joule

Now, $W_2 = s4\pi R^2(4-1) = 4\pi R^2 s \times 3$

$\Rightarrow \quad W_2 = 3 \times \frac{10}{2} = 15$ Joule

31. The Young's double slit interference experiment is performed using light consisting of 480 nm and 600 nm wavelengths to form interference patterns. The least number of the bright fringes of 480 nm light that are required for the first coincidence with the bright fringes formed by 600 nm light is

(1) 6

(2) 8

(3) 5

(4) 4

Answer (3)

Sol. Fringe width of λ_1 = 480 nm

$$\Delta w_1 = \frac{\lambda_1 D}{d} = 480\left(\frac{D}{d}\right)$$

Similarly fringe width for λ_2 = 600 nm

$$\Delta w_2 = \lambda_2\left(\frac{D}{d}\right) = 600\left(\frac{D}{d}\right)$$

Clearly we can see that for minimum value

$5(\Delta w_1) = 4(\Delta w_2)$

That mean 5th maxima of λ_1 = 480 nm

Coincides with 4th maxima of λ_2 = 600 nm



32. An ideal gas goes from an initial state to final state. During the process, the pressure of gas increases linearly with temperature.

A. The work done by gas during the process is zero.

B. The heat added to gas is different from change in its internal energy.

C. The volume of the gas is increased.

D. The internal energy of the gas is increased.

E. The process is isochoric (constant volume process)

Choose the **correct** answer from the options given below:

(1) A, D, E only (2) E only

(3) A, C only (4) A, B, C, D only

Answer (1)

Sol. Given for ideal gas.

P = constant T

$\Rightarrow PT^{-1}$ = constant

$\Rightarrow \frac{P}{PV}$ = constant = V is constant

So, isochoric process

Work done by gas will be zero.

A, D, E only correct.

33. A car of mass 'm' moves on a banked road having radius 'r' and banking angle θ. To avoid slipping from banked road, the maximum permissible speed of the car is v_0. The coefficient of friction μ between the wheels of the car and the banked road is

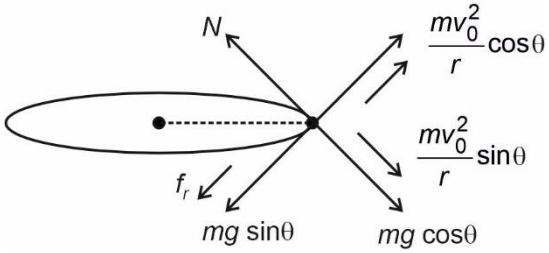
(1) $\mu = \frac{v_0^2 + rg\tan\theta}{rg - v_0^2\tan\theta}$ (2) $\mu = \frac{v_0^2 + rg\tan\theta}{rg + v_0^2\tan\theta}$

(3) $\mu = \frac{v_0^2 - rg\tan\theta}{rg + v_0^2\tan\theta}$ (4) $\mu = \frac{v_0^2 - rg\tan\theta}{rg - v_0^2\tan\theta}$

Answer (3)

Sol. Given maximum possible speed = v_0

For this



So, $N = mg\cos\theta + \frac{mv_0^2}{r}\sin\theta$

$f_r = \mu mg\cos\theta + \frac{\mu mv_0^2}{r}\sin\theta$

And $\frac{mv_0^2}{r}\cos\theta = mg\sin\theta + f_r$

$\Rightarrow \frac{mv_0^2}{r}\cos\theta - mg\sin\theta = \mu\left(mg\cos\theta + \frac{mv_0^2}{r}\sin\theta\right)$

$\Rightarrow \left(v_0^2 - gr\tan\theta\right) = \mu\left(v_0^2\tan\theta + gr\right)$

$\Rightarrow \mu = \frac{v_0^2 - gr\tan\theta}{gr + v_0^2\tan\theta}$

34. An air bubble of radius 0.1 cm lies at a depth of 20 cm below the free surface of a liquid of density 1000 kg/m^3. If the pressure inside the bubble is 2100 N/m^2 greater than the atmospheric pressure, then the surface tension of the liquid in SI unit is (use g = 10 m/s^2)

(1) 0.02

(2) 0.05

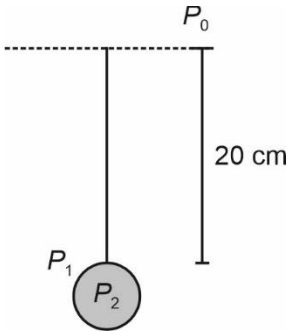
(3) 0.1

(4) 0.25

Answer (2)



Sol.



$$P_1 = P_0 + \rho gh = P_0 + 1000 \times 10 \times \frac{20}{100}$$

$$\Rightarrow \quad P_1 = P_0 + 2000$$

So, $P_2 - P_1 = \frac{2S}{R} = \left(\frac{2S}{1\times10^{-3}}\right)$

$\Rightarrow \quad P_2 = P_0 + 2100$ (given)

So, $P_0 + 2100 - P_0 - 2000 = 2S \times 10^3$

$\Rightarrow \quad 100 = 2S \times 10^3$

$$\Rightarrow \quad s = \left(\frac{1}{20}\right) = 0.05$$

35. A parallel plate capacitor was made with two rectangular plates, each with a length of l = 3 cm and breadth of b = 1 cm. The distance between the plates is 3 μm. Out of the following, which are the ways to increase the capacitance by a factor of 10?

A. l = 30 cm, b = 1 cm, d = 1 μm

B. l = 3 cm, b = 1 cm, d = 30 μm

C. l = 6 cm, b = 5 cm, d = 3 μm

D. l = 1 cm, b = 1 cm, d = 10 μm

E. l = 5 cm, b = 2 cm, d = 1 μm

Choose the correct answer from the options given below:

(1) C and E only

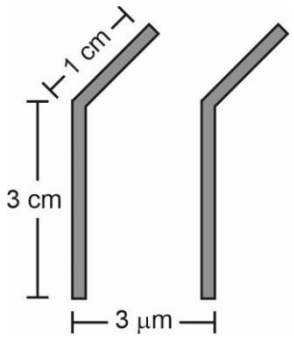
(2) C only

(3) B and D only

(4) A only

Answer (1)

Sol.



We know $C = \frac{A\varepsilon_0}{d}$

$$= \frac{b\ell\varepsilon_0}{d}$$

So to increase the capacitance by 10 factor $\left(\frac{A}{d}\right)$ has to increase by 10 fator.

For option (A) $C' = \frac{(30\ell)\, b\varepsilon_0}{\left(\frac{d}{3}\right)} = 30C$

For option (B) $C' = \frac{\ell\, b\varepsilon_0}{10d} = \frac{C}{10}$

For option (C) $C' = \frac{(2\ell)5b\varepsilon_0}{d} = 10C$

For option (D) $C' = \frac{\left(\frac{\ell}{3}\right)b\varepsilon_0}{\left(\frac{10d}{3}\right)} = \frac{C}{10}$

For option (E) $C' = \frac{\left(\frac{\ell}{3}\right)5(2b)\varepsilon_0}{\left(\frac{d}{3}\right)} = 10C$

Clearly (C) and (E) are the situation for 10C

36. A uniform solid cylinder of mass 'm' and radius 'r' rolls along an inclined rough plane of inclination 45°. If it starts to roll from rest from the top of the plane then the linear acceleration of the cylinder's axis will be

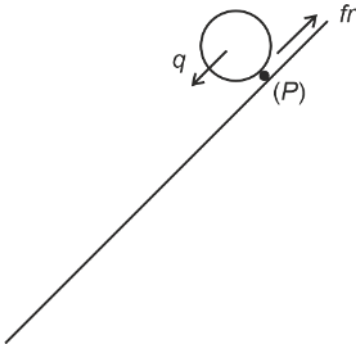
(1) $\frac{1}{3\sqrt{2}}g$ (2) $\frac{\sqrt{2}g}{3}$

(3) $\sqrt{2}\,g$ (4) $\frac{1}{\sqrt{2}}g$

Answer (2)



Sol.



For pure rolling about point '*P*'.

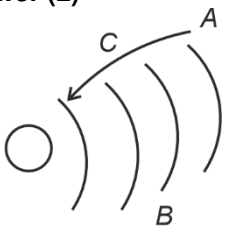
$$\Rightarrow \quad a = \frac{2g}{3}\sin\theta = \frac{2g}{3\sqrt{2}} = \frac{\sqrt{2}}{3}g$$

37 During the transition of electron from state *A* to state *C* of a Bohr atom, the wavelength of emitted radiation is 2000 Å and it becomes 6000 Å when the electron jumps from state *B* to state *C*. Then the wavelength of the radiation emitted during the transition of electrons from state *A* to state *B* is

(1) 2000 Å (2) 3000 Å
(3) 6000 Å (4) 4000 Å

Answer (2)

Sol.



For *A* to *C*

$$\frac{hc}{\lambda_1} = E_0 z^2\left(\frac{1}{n_c^2} - \frac{1}{n_A^2}\right) \quad \ldots(i)$$

And $\frac{hc}{\lambda_2} = E_0 z^2\left(\frac{1}{n_c^2} - \frac{1}{n_B^2}\right)$ ...(ii)

So for *A* and *B*

$$\frac{hc}{\lambda_3} = E_0 z^2\left(\frac{1}{n_B^2} - \frac{1}{n_A^2}\right)$$

Clearly subtracting equation (ii) from equation (i)

$$hc\left[\frac{1}{\lambda_1} - \frac{1}{\lambda_2}\right] = E_0 z^2\left[\frac{1}{n_B^2} - \frac{1}{n_A^2}\right] = \frac{hc}{\lambda_3}$$

$$\Rightarrow \quad \frac{1}{\lambda_3} = \frac{1}{\lambda_1} - \frac{1}{\lambda_2} \Rightarrow \frac{1}{\lambda_3} = \frac{(6000-2000)}{6000\times 2000} = \frac{1}{3000}$$

$\lambda_3 = 3000$Å

38. An object of mass '*m*' is projected from origin in a vertical *xy* plane at an angle 45° with the *x*- axis with an initial velocity v_0. The magnitude and direction of the angular momentum of the object with respect to origin, when it reaches at the maximum height, will be [*g* is acceleration due to gravity]

(1) $\frac{mv_0^3}{4\sqrt{2}g}$ along positive z-axis

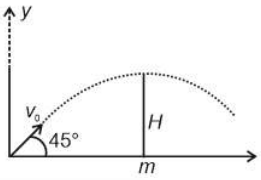
(2) $\frac{mv_0^3}{2\sqrt{2}g}$ along negative z-axis

(3) $\frac{mv_0^3}{2\sqrt{2}g}$ along positive z-axis

(4) $\frac{mv_0^3}{4\sqrt{2}g}$ along negative z-axis

Answer (4)

Sol.



At height point

The speed is $v_x = v\cos 45°$

$$v_x = \frac{v_0}{\sqrt{2}}$$

So angular momentum about origin is

$L = mv_x \cdot H$

$$\Rightarrow \quad L = m\frac{v_0}{\sqrt{2}}\cdot\left(\frac{v_0}{\sqrt{2}}\right)^2 \times \frac{1}{2g}$$

$$\Rightarrow \quad L = \frac{mv_0^3}{4\sqrt{2}g} \ (-\mu\text{- } z\text{-axis})$$

39. Consider the following statements:

A. The junction area of solar cell is made very narrow compared to a photo diode.
B. Solar cells are not connected with any external bias.
C. LED is made of lightly doped p-n junction.
D. Increase of forward current results in continuous increase of LED light intensity.
E. LEDs have to be connected in forward bias for emission of light.

Choose the **correct** answer from the options given below:

(1) B, E Only (2) B, D, E Only
(3) A, C, E Only (4) A, C Only

Answer (1)



Sol. Clearly statement 'A' is wrong as for solar cell junction area are mode wide. LED connected in forward biase and its intensity increases upto certain value of current and further there is no charge as intensity saturates.

Also solar cells are not connected with any external bias. So '*B*' is also correct

40. What is the relative decrease in focal length of a lens for an increase in optical power by 0.1D from 2.5 D? ['D' stands for dioptre]

(1) 0.01 (2) 0.04

(3) 0.1 (4) 0.40

Answer (2)

Sol. $P = \frac{1}{f}$

So $P_1 = \frac{1}{F_1} \Rightarrow F_1 = \frac{1}{2.5}$

Next $F_2 = \frac{1}{2.6}$

$$\left|\frac{\Delta F}{F_1}\right| = \frac{|F_2 - F_1|}{F_1} = \left|\frac{F_2}{F_1} - 1\right|$$

$$\Rightarrow \left|\frac{\Delta F}{F_1}\right| = \left|\frac{2.5}{2.6} - 1\right| = \frac{0.1}{2.6} \approx 0.04$$

41. A force $F = \alpha + \beta x^2$ acts on an object in the x-direction. The work done by the force is 5 J when the object is displaced by 1 m. If the constant α = 1N then β will be

(1) 8 N/m^2 (2) 15 N/m^2

(3) 12 N/m^2 (4) 10 N/m^2

Answer (3)

Sol. $F = \alpha + \beta x^2$

Work done $\int dw = \int F \cdot dx$

$$\Rightarrow \Delta w = \int F \cdot dx = \int \left(\alpha + \beta x^2\right) dx$$

$$\Rightarrow \Delta w = \left|\alpha x + \frac{\beta x^3}{3}\right|_0^1 = \alpha + \frac{\beta}{3} = 5$$

Given α = 1

So, $\frac{\beta}{3} = 4$

$$\Rightarrow \beta = 12 \text{ N/m}^2$$

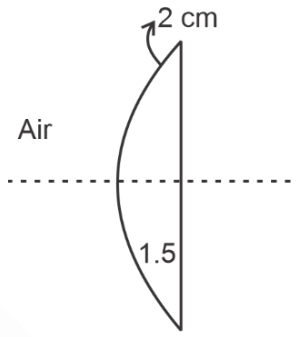
42. A plano-convex lens having radius of curvature of first surface 2 cm exhibits focal length of f_1 in air. Another plano-convex lens with first surface radius of curvature 3 cm has focal length of f_2 when it is immersed in a liquid of refractive index 1.2. If both the lenses are made of same glass of refractive index 1.5, the ratio of f_1 and f_2 will be

(1) 1 : 2 (2) 1 : 3

(3) 2 : 3 (4) 3 : 5

Answer (2)

Sol. For case (1)



$$\frac{1.5}{v_1} - \frac{1}{\infty} = \frac{(1.5-1)}{R_1}$$

$$\frac{1}{f_1} - \frac{1.5}{v_1} = \frac{1-1.5}{\infty}$$

$$\Rightarrow \frac{1}{f_1} = \frac{(1.5-1)}{R_1} \quad \ldots(1)$$

$$\Rightarrow \frac{1}{f_1} = \frac{1}{2R_1}$$

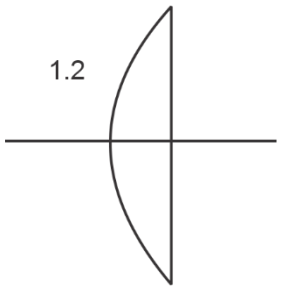
For case (2)

$$\frac{1.5}{v_1} - \frac{1.2}{\infty} = \frac{(1.5-1.2)}{R_2}$$

$$\frac{1.2}{f_2} - \frac{1.5}{v_1} = \frac{(1.2-1.5)}{\infty}$$

$$\Rightarrow \frac{1.2}{f_2} = \frac{0.3}{R_2}$$

$$\Rightarrow \frac{1}{f_2} = \frac{1}{4R_2}$$



So clearly, $\frac{f_1}{f_2} = \frac{2 \times 2}{4 \times 3} = \frac{1}{3}$



43. A particle is executing simple harmonic motion with time period 2 s and amplitude 1 cm. If D and d are the total distance and displacement covered by the particle in 12.5 s, then $\frac{D}{d}$ is

(1) 10 (2) $\frac{15}{4}$

(3) $\frac{16}{5}$ (4) 25

Answer (4)

Sol. $T = 2$ s

$A = 1$ cm



In 12.5 s 6 total oscillation and 1 quarter oscillation.

So, $D = (6 \times 4A) + A = 25A$

$d = A$

$\frac{D}{d} = 25$

44. An alternating current is given by $I = I_A \sin\omega t + I_B \cos\omega t$. The r.m.s current will be

(1) $\sqrt{\frac{I_A^2 + I_B^2}{2}}$ (2) $\frac{\sqrt{I_A^2 + I_B^2}}{2}$

(3) $\sqrt{I_A^2 + I_B^2}$ (4) $\frac{|I_A + I_B|}{\sqrt{2}}$

Answer (1)

Sol. $I = I_A \sin\omega t + I_B \cos\omega t$

$I_{rms} = \sqrt{I_A^2 + I_B^2}$

So, $I_{RMS} = \sqrt{\frac{I_A^2 + I_B^2}{2}}$

45. An electron of mass 'm' with an initial velocity $\vec{v} = v_0\hat{i}\,(v_0 > 0)$ enters an electric field $\vec{E} = -E_0\hat{k}$. If the initial de Broglie wavelength is λ_0, the value after t would be

(1) $\frac{\lambda_0}{\sqrt{1 + \frac{e^2E_0^2t^2}{m^2v_0^2}}}$ (2) $\lambda_0\sqrt{1 + \frac{e^2E_0^2t^2}{m^2v_0^2}}$

(3) λ_0 (4) $\frac{\lambda_0}{\sqrt{1 - \frac{e^2E_0^2t^2}{m^2v_0^2}}}$

Answer (1)

Sol. $mv_0 \longrightarrow \otimes$

$E = -E_0\hat{k}$

$\lambda_0 = \frac{h}{mv_0}$

After time 't' $v_{(t)} = \left(\frac{E_0e}{m}\right)\cdot t$

So, $v = \sqrt{v_0^2 + \left(\frac{eE_0t}{m}\right)^2}$

Now, $\lambda = \frac{h}{m\sqrt{v_0^2 + \left(\frac{eE_0t}{m}\right)^2}}$

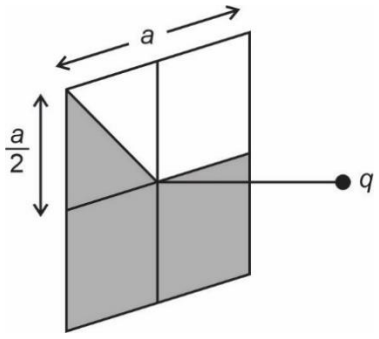
$\lambda_0 = \frac{h}{mv_0}$

$\Rightarrow \frac{\lambda}{\lambda_0} = \frac{1}{\sqrt{1 + \left(\frac{eE_0t}{mv_0}\right)^2}}$

SECTION - B

Numerical Value Type Questions: This section contains 5 Numerical based questions. The answer to each question should be rounded-off to the nearest integer.

46. A square loop of side $a = 1$ m is held normally in front of a point charge $q = 1$C. The flux of the electric field through the shaded region is $\frac{5}{p} \times \frac{1}{\varepsilon_0}\frac{\text{Nm}^2}{\text{C}}$, where the value of p is _____.



Answer (48)



Sol. Flux through complete surface

$$\phi_1 = \frac{q}{\varepsilon_0 \times 6}$$

So flux through shaded region is

$$\phi_2 = \phi_1 \frac{5}{8} = \frac{q}{6\varepsilon_0} \times \frac{5}{8} = \frac{5q}{48\varepsilon_0}$$

So, $p = 48$

47. The least count of a screw gauge is 0.01 mm. If the pitch is increased by 75% and number of divisions on the circular scale is reduced by 50%, the new least count will be _____ $\times 10^{-3}$ mm.

Answer (35)

Sol. Least count $= 0.01 = \left(\frac{L}{N}\right)$

Now $\frac{1.75L}{N/2}$ = New least count

$$\Rightarrow 3.5 \times \frac{L}{N} = 3.5 \times 0.01 \text{ mm}$$

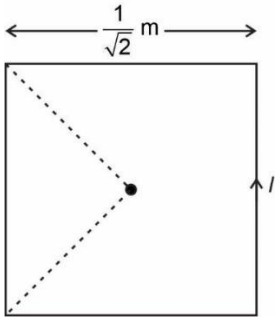
$$= 35 \times 10^{-3} \text{ mm}$$

48. A current of 5 A exists in a square loop of side $\frac{1}{\sqrt{2}}$ m. Then the magnitude of the magnetic field B at the centre of the square loop will be $p \times 10^{-6}$ T. Where, value of p is _____.

[Take $\mu_0 = 4\pi \times 10^{-7}$ T m A^{-1}].

Answer (8)

Sol.



$$B = \frac{\mu_0 I \times 2}{4\pi \ell} \times 2\cos 45° \times 4$$

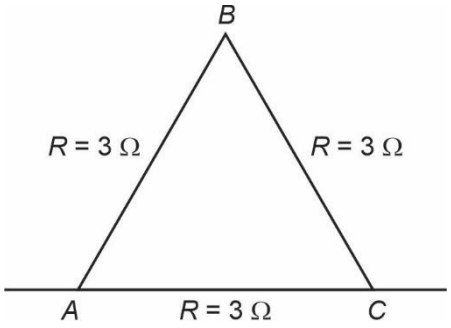
$$\Rightarrow B = \frac{4\pi \times 10^{-7} \times \sqrt{2}}{4\pi \times 1} \times 4 \times 4 \times \frac{1}{\sqrt{2}} \times 5$$

$$\Rightarrow B = 8 \times 10^{-6} \text{ T}$$

49. A wire of resistance 9 Ω is bent to form an equilateral triangle. Then the equivalent resistance across any two vertices will be _____ ohm.

Answer (2)

Sol.



Equivalent resistance across AC is

$$R_1 = 2R \,||\, R = \frac{2R \times R}{3R} = \frac{2R}{3}$$

$$\Rightarrow R_1 = 2\ \Omega$$

50. The temperature of 1 mole of an ideal monoatomic gas is increased by 50°C at constant pressure. The total heat added the change in internal energy are E_1 and E_2, respectively. If $\frac{E_1}{E_2} = \frac{x}{9}$ then the value of x is _____.

Answer (15)

Sol. $\Delta T = 50°C = 50$ K

$N = 1$ mole ideal gas

$\Delta P = 0$

$$\Delta Q_P = NC_P\Delta T = E_1 = N \cdot \frac{5}{2} R\Delta T$$

$$\Delta Q_V = NC_V\Delta T = E_2 = N\frac{3}{2} R\Delta T$$

$$\frac{E_1}{E_2} = \frac{x}{9} = \frac{C_P}{C_V}$$

For monoatomic gas = $\frac{C_P}{C_V} = \frac{5}{3}$

$$\Rightarrow \frac{x}{9} = \frac{5}{3}$$

$$\Rightarrow x = 15$$

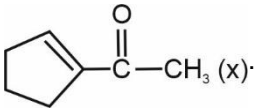


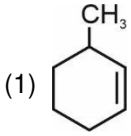
CHEMISTRY

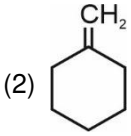
SECTION - A

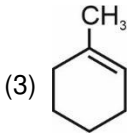
Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

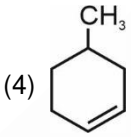
Choose the correct answer :

51. Aman has been asked to synthesise the molecule  (x). He thought of preparing the molecule using an aldol condensation reaction. He found a few cyclic alkenes in his laboratory. He thought of performing ozonolysis reaction on alkene to produce a dicarbonyl compound followed by aldol reaction to prepare "x". Predict the suitable alkene that can lead to the formation of "x".

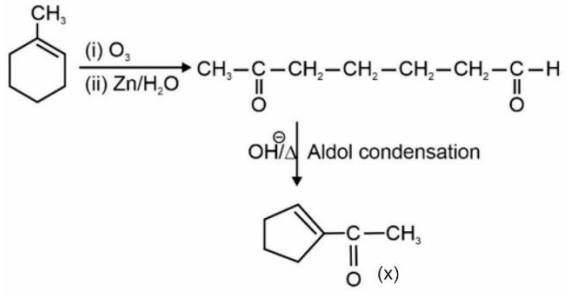
(1) 

(2) 

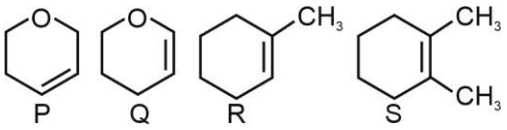
(3) 

(4) 

Answer (3)

Sol. 

52. Following are the four molecules "P", "Q", "R" and "S". Which one among the four molecules will react with H–Br(aq) at the fastest rate?

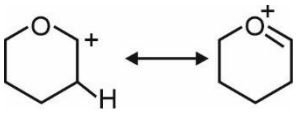


(1) P (2) R

(3) Q (4) S

Answer (3)

Sol. The reaction with H — Br will be fastest for the specie forming most stable carbocation



Due to resonance, it will have most stable carbocation.

53. One mole of the octahedral complex compound $Co(NH_3)_5Cl_3$ gives 3 moles of ions on dissolution in water. One mole of the same complex reacts with excess of $AgNO_3$ solution to yield two moles of AgCl(s). The structure of the complex is

(1) $[Co(NH_3)_4Cl].Cl_2.NH_3$

(2) $[Co(NH_3)_4Cl_2].Cl.NH_3$

(3) $[Co(NH_3)_3Cl_3].2NH_3$

(4) $[Co(NH_3)_5Cl]Cl_2$

Answer (4)

Sol. $[Co(NH_3)_5Cl]Cl_2 \xrightarrow{H_2O} \underbrace{[Co(NH_3)_5Cl]^{2+} + 2Cl^-}_{3 \text{ ions}}$

$\downarrow + AgNO_3$

2 mole AgCl

54. The carbohydrate "Ribose" present in DNA, is

A. A pentose sugar

B. present in pyranose form

C. in "D" configuration

D. a reducing sugar, when free

E. in α – anomeric form

Choose the correct answer from the options given below

(1) A, B and E only (2) B, D and E only

(3) A, C and D only (4) A, D and E only

Answer (3)

